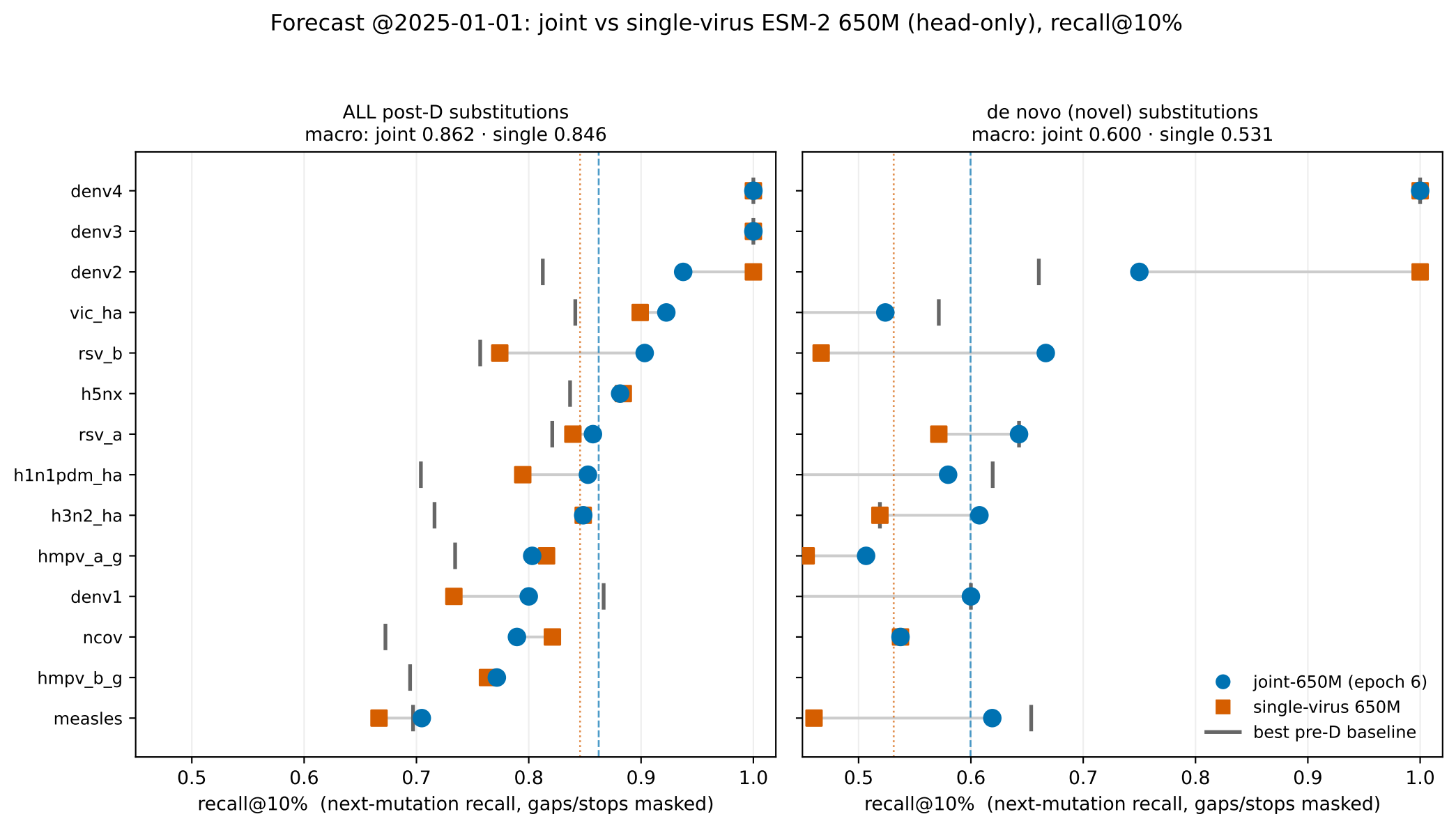
Forecast @2025-01-01: joint vs single-virus ESM-2 650M (head-only), recall@10%
ALL post-D substitutions
macro: joint 0.862 · single 0.846
de novo (novel) substitutions
macro: joint 0.600 · single 0.531
denv4
denv3
denv2
vic_ha
rsv_b
h5nx
rsv_a
h1n1pdm_ha
h3n2_ha
hmpv_a_g
denv1
ncov
hmpv_b_g
measles
0.5
0.6
0.7
0.8
0.9
1.0
recall@10% (next-mutation recall, gaps/stops masked)
joint-650M (epoch 6)
single-virus 650M
best pre-D baseline